B2-base: Constraint-preserving quarter-hourly price reconstruction

AET | Neural Wave 2026 - concise methodology and results report

The problem. For every delivery date and each of four anonymised market zones, the pipeline receives a complete 24-hour day-ahead curve and must reconstruct 96 quarter-hour prices. The hourly average is not a prediction target: it is a hard constraint. Our model therefore learns only the intra-hour shape, using the official data package and history from strictly earlier delivery dates.

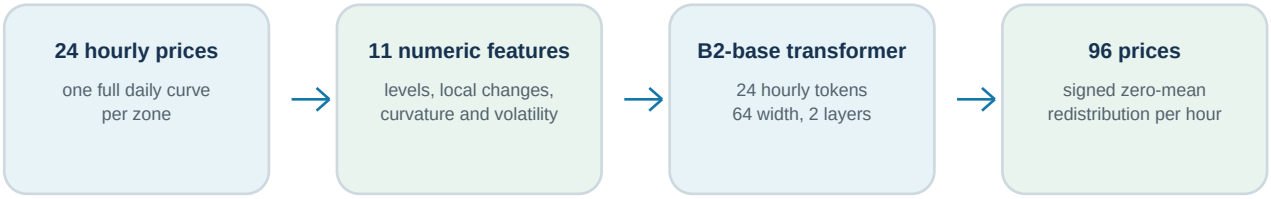

Figure 1. Frozen B2-base inference path. No quarter-hour price from the delivery date is used as an input.

Approach

B2-base is a compact daily Transformer. It processes 24 hourly tokens jointly, so each hour can use the context of the full known hourly curve. The model uses 11 target-free numerical features derived from that curve: normalized level, left/right changes, curvature, daily location and scale, and local volatility. Zone conditioning and Fourier hour features are supplied separately; no external data or pretrained weights are used.

Its signed redistribution head produces a four-value correction with exact zero mean. Predictions are formed as the supplied hourly price plus that correction. This parameterisation makes the four-quarter mean equal to the supplied hourly price by construction, rather than relying on a penalty or a post-hoc approximation. The base backbone has width 64, two Transformer layers, four attention heads and a 128-unit feed-forward block (135,113 trainable parameters).

Training uses PyTorch and Lightning with MAE loss, AdamW, dropout 0.1 and per-day normalization. The final reported B2-base transfer run is a fixed 30-epoch refit on November 2025-September 2026; October 2025 is scored once as an already-inspected transfer diagnostic. At challenge inference, the frozen code can read earlier quarter-hour history only through rules fixed before evaluation.

Full-day context All 24 supplied hourly values are available at once, so the reconstruction can respond to ramps and daily level.	**Exact by construction** The zero-mean correction preserves every hourly average without a numerical compromise.	**Small enough to audit** A direct 24-token design avoids hidden fallback rules and unnecessary moving parts.

Results and validation

The official metric averages daily quarter-hour MAE within each zone and then gives the four zones equal weight. We retained the user-selected B2-base model because it gave the strongest observed fixed-30 October score in the B2 family. This is a transfer result, not a license to tune on October; the validation protocol remained separate.

4.318	0.864	16.7%	5.0e-11
October transfer MAE EUR/MWh	absolute gain vs MAKIMA EUR/MWh	relative gain vs MAKIMA (October, derived)	maximum hourly-mean error EUR/MWh

The standalone fixed-amplitude MAKIMA reference scored approximately 5.182 MAE on the same October comparison (4.318 + 0.864). The flat baseline repeats each hourly input four times; it is maintained as the challenge comparison baseline. All serialized predictions were independently audited: 480 files from 160 completed fits had a maximum hourly-mean mismatch of 5.00e-11 EUR/MWh, comfortably below the 1e-4 eligibility tolerance.

Validation design

Model development used HORIZON: eight expanding monthly validation folds from February through September 2026, while October stayed out of selection. Each fold fits preprocessing only on its own training dates and uses fixed calendar-day validation windows. Best checkpoints are selected by validation MAE (maximum 50 epochs, patience 10); HARVEST refits do not select on October. The reported B2-base HORIZON score is 4.292 macro-month MAE (4.300 pooled-date MAE), with all 8/8 folds present.

What we tried

Family	Core idea	What it taught us
Flat / MAKIMA	Repeat hourly price; then smooth, zero-mean interpolation.	A reliable constraint-preserving reference; learned models must earn their added complexity.
Gradient boosting	XGBoost, LightGBM and CatBoost on engineered hourly features.	Strong tabular baseline, but a different historical split and feature pipeline made it unsuitable as the final frozen choice.
GANN / MLP	Compact residual and gradient-based neural models.	Useful early baselines; later daily Transformer experiments improved the transfer result.
A-family Transformers	Masked 96-token models with MAKIMA, teacher revelation and a training-only gate.	Competitive, but more moving parts. B2-base was preferred for the final simple signed head path.
B0/B1/B3 and scaling	MLP vs Transformer, 11 vs 19 features, MAKIMA prior, and four backbone sizes.	Larger models improved some validation means but did not transfer consistently; B2-base was best observed B2 fixed-30 transfer run.

Table 1. Comparisons are descriptive unless evaluated under their stated validation protocol; scores from different historical split designs are not used as a common ranking.

Selection discipline
HORIZON governs model development. October is a separate, already-inspected transfer check and never selects a checkpoint, gate or hyperparameter.

What the final number means
The 4.318 MAE is the official daily, equal-zone aggregation for the fixed-30 B2-base October refit - a transparent reference point, not a future-score claim.

Challenges, safeguards and reproducibility

Exact consistency. A conventional unconstrained 96-output regression can be accurate but fail eligibility because its quarters do not average exactly to the supplied hour. The signed redistribution head solves this structurally: it subtracts two softmax distributions, scales the difference, and re-centres it. The output layer therefore has unrestricted shape but zero hourly mean. We also validate the final serialized file before scoring.

Time leakage and seasonal drift. The most tempting feature - the target day's quarter-hour profile - is forbidden. All learned statistics are fit only on original training dates; inference rules consume only strictly earlier quarter-hour history. We preserve CET/CEST delivery semantics and omit the two daylight-saving transition days. Chronological HORIZON folds expose a material month-to-month MAE range, so reported performance is not presented as a stationary-data guarantee.

Complexity that did not pay reliably. The scaling study found modest validation gains from larger Transformers, but the added capacity did not consistently improve October transfer. Likewise, a MAKIMA prior, masked-token teacher revelation and gated output heads increased pipeline complexity. They remain documented experiments, not hidden components of B2-base. The final selected model is deliberately smaller and uses a direct hourly-token representation.

Reproducible delivery

What is frozen	How it is checked
Inputs	Only test_hourly.csv and prior history_15min.csv; no network access, external data, target-day quarter-hour inputs or evaluation-time retraining.
Training	Seeded PyTorch/Lightning configuration, locked dependencies, saved resolved configuration, source snapshot and local checkpoints. Long neural runs log metadata and metrics without uploading checkpoints.
Output	One finite prediction for every delivery date, zone, hour and quarter; duplicate/missing-key checks and the hourly-mean tolerance are validated before submission.
Entry points	Configuration-driven Lightning training and a documented inference pipeline are kept in the repository so the jury can retrain from the official package and run predictions without manual edits.

1. Keys	2. Values	3. Constraint	4. Provenance
One row for each date, zone, hour and quarter.	Four finite numeric quarter-hour predictions per hour.	Hourly-average mismatch must remain below 1e-4 EUR/MWh.	Frozen code, configuration, seed and artefacts travel together.

AI assistants and LLM usage

We used an LLM-based coding assistant as a development tool for code navigation, implementation support, test assistance, documentation and this report's editing. It did not supply external market data, pretrained weights or evaluation targets. Team members reviewed the methodology, selected the submitted B2-base model, executed experiments, and verified the final results and constraint checks. All model fitting and evaluation described here use the official challenge data package only.

References and evidence

Challenge rules: *AET | Neural Wave 2026 Challenge*, supplied by the organisers. Experiment evidence: repository configurations and recorded outputs for the controlled matrix, HORIZON/HARVEST validation, scaling study and replication study. Reported MAEs are EUR/MWh and use the required daily, equal-zone aggregation unless explicitly labelled otherwise.